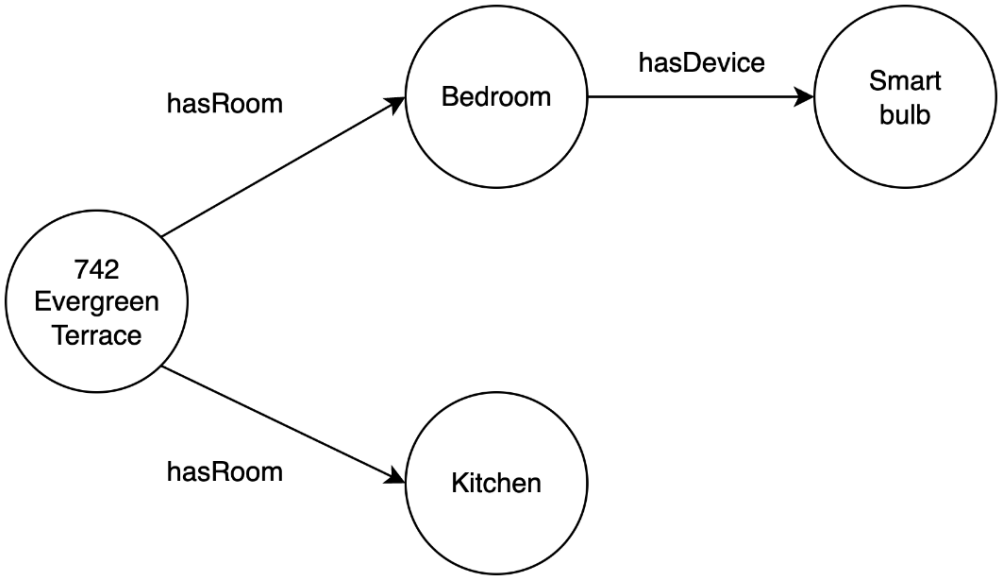



**Figure 4.7 A house, its rooms, and their devices modelled as a graph structure.**

### 4.3.4 Document structures

Document structures store data as flexible, schema-less documents that can contain nested structures and variable attributes. This approach works well for semi-structured data from APIs, configuration files, or sensor data where different device types report different sets of measurements. A simple example of how we might model a house, it's rooms, and their devices as a document record is:

```
{"id": "_496756956732423",
  "name": "742 Evergreen Terrace"
  "rooms": [
    { "name": "Bedroom",
      "area": 10,
      "devices": [
        { "name": "Smart bulb"
          "address": "00:1B:63:84:45:E6"
        }
      ]
    },
    { "name": "Kitchen",
      "area": 20
    }
  ]
}
```

Document structures provide schema flexibility that accommodates evolving data requirements without requiring database migrations. However, this flexibility can complicate queries that span multiple documents or require joins across different data types.